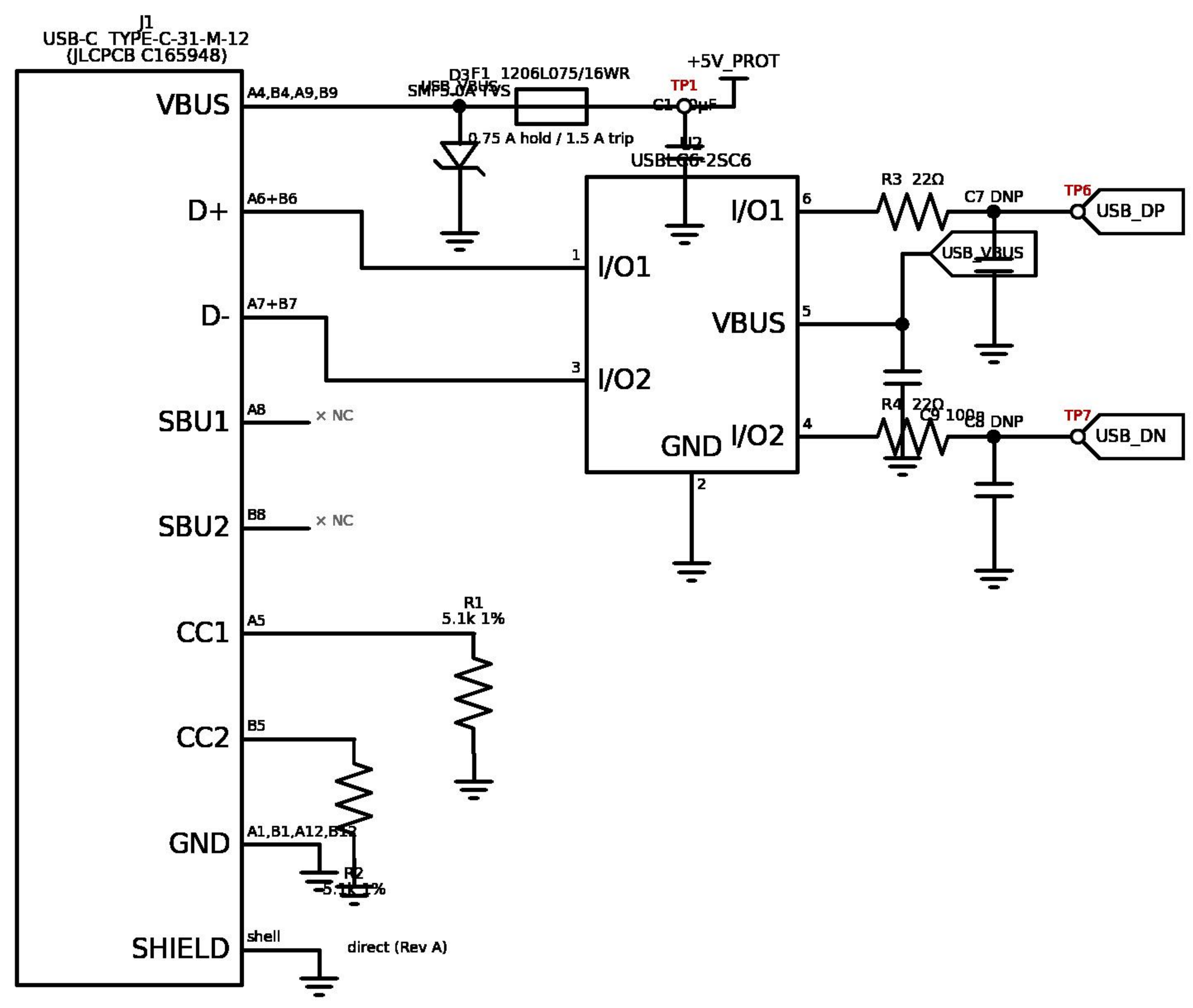
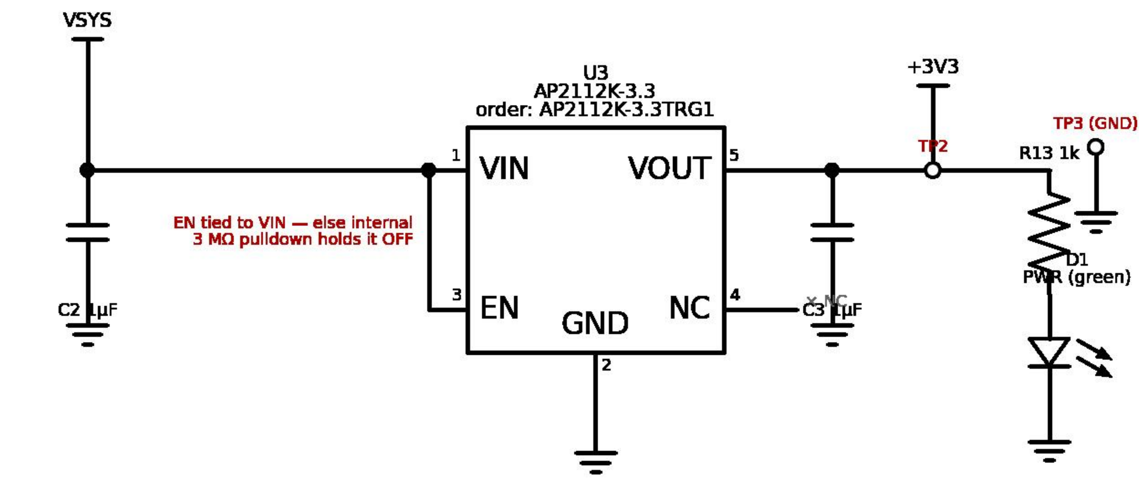
ESP32-S3 PLANT MONITOR — COMPLETE REFERENCE SCHEMATIC — REV A.2

02 — USB-C INPUT & PROTECTION



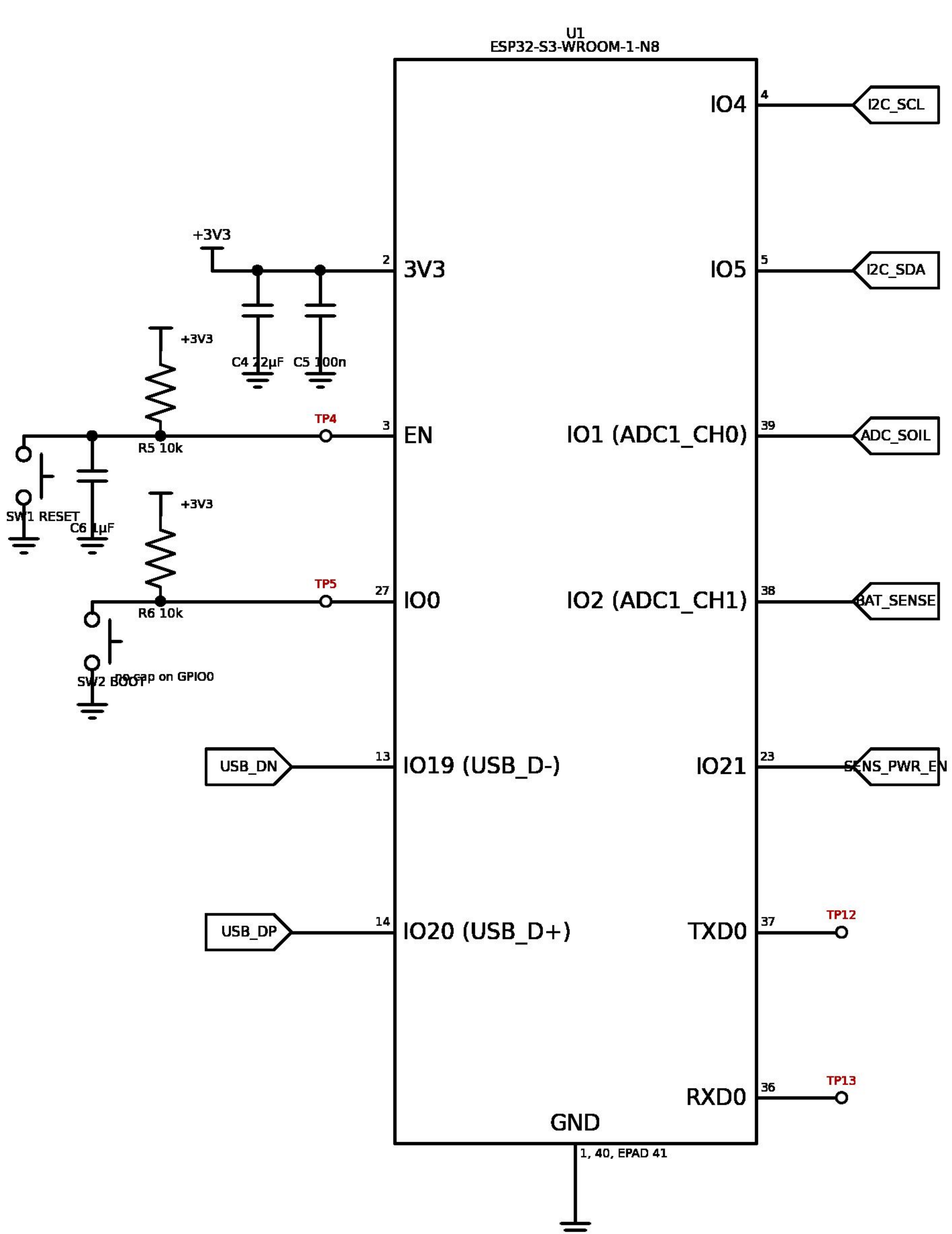
Notes: A6+B6 and A7+B7 tied at the connector (16-pin footprint has separate pads). One 5.1 kΩ per CC pin = USB-C default-power sink (TA0357 Tbl 3); never join CC1/CC2. D3 clamps VBUS surges at the entrance (HDG §1.3.2); C1 = power-entrance bulk. U2 is flow-through (in 1/3, out 6/4) — place D3, F1, U2 at the connector. R3/R4 placed near the module. C7/C8 = reserved footprints, Do Not Populate. F1: hold derates to 0.65 A at 40 °C — sized for Wi-Fi peak + 100 mA charging simultaneously.

03 — 3.3 V REGULATOR (VSYS → +3V3)



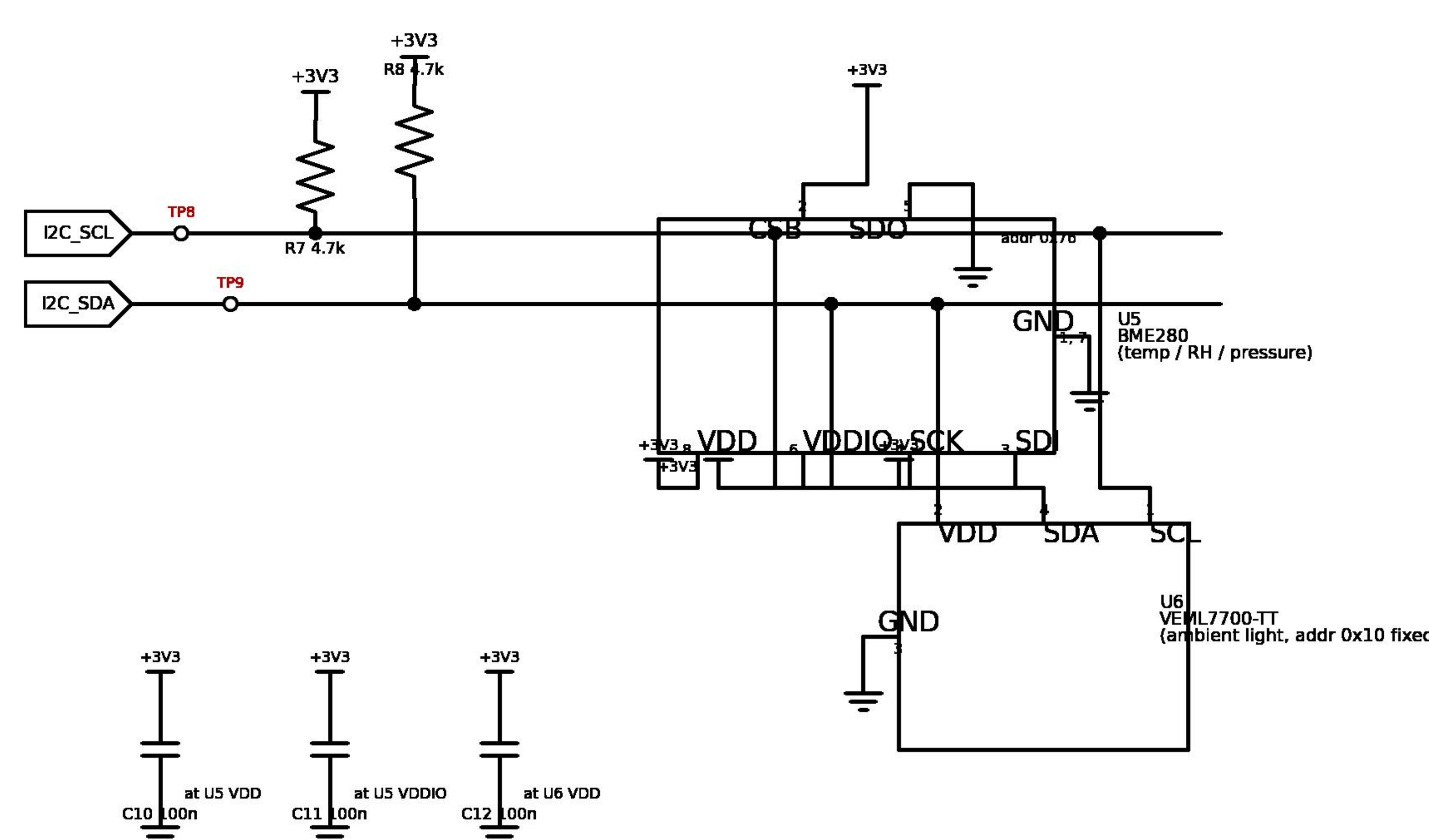
Notes: Input VSYS = ≈4.6 V on USB, or the battery (3.0–4.2 V) when unplugged. Output 3.3 V, 600 mA capability; dropout ≈250 mV → below ≈3.6 V battery the rail sags gracefully with the battery (accepted Rev A decision; the battery monitor reports it). C2/C3 = datasheet 1 µF ceramic. D1 may be DNP for battery-life tests. Thermal: ≈0.34 W sustained worst case — give U3 generous copper.

04 — ESP32-S3 CORE



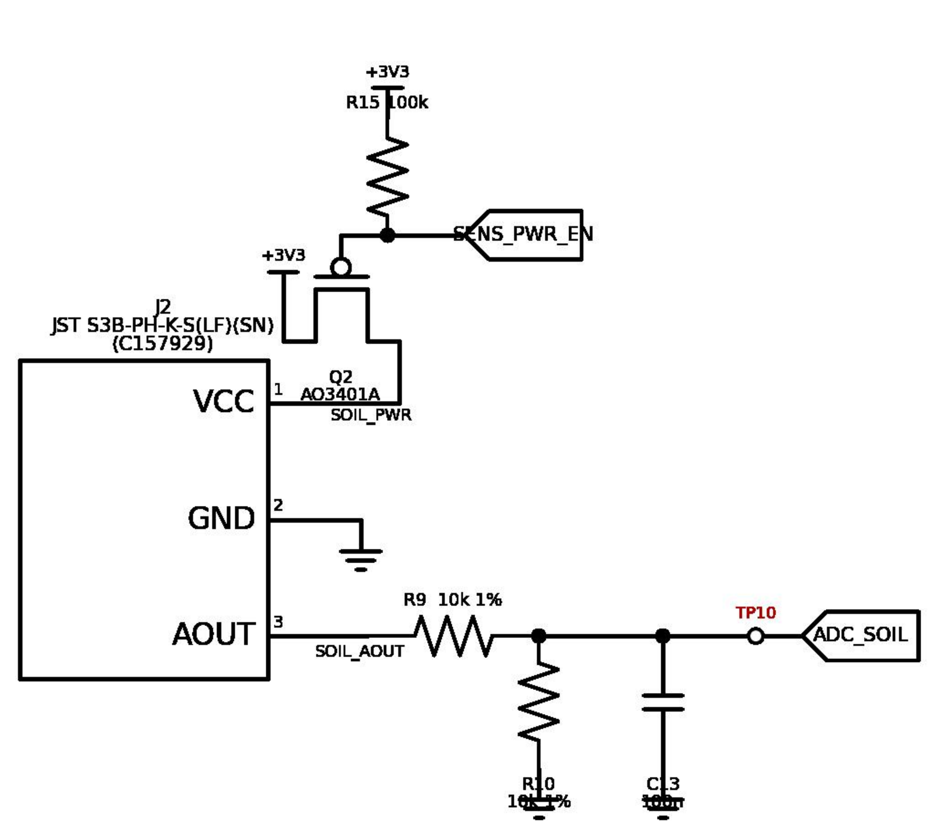
Strapping pins (HDG §1.3.9): GPIO0, GPIO3, GPIO45, GPIO46. GPIO3 / GPIO45 / GPIO46: leave UNCONNECTED — internal defaults are correct; never attach anything that loads them at boot. No large capacitor on GPIO0. IO2 reads half the battery voltage (ADC1_CH1); IO21 drives the soil-probe power switch (low = probe ON; R15 keeps it OFF through boot). All other GPIO left unconnected in Rev A. TXD0/RXD0 test points = UART recovery/logging fallback if native USB is ever unusable (ROM supports UART download boot).

05 — I2C SENSORS



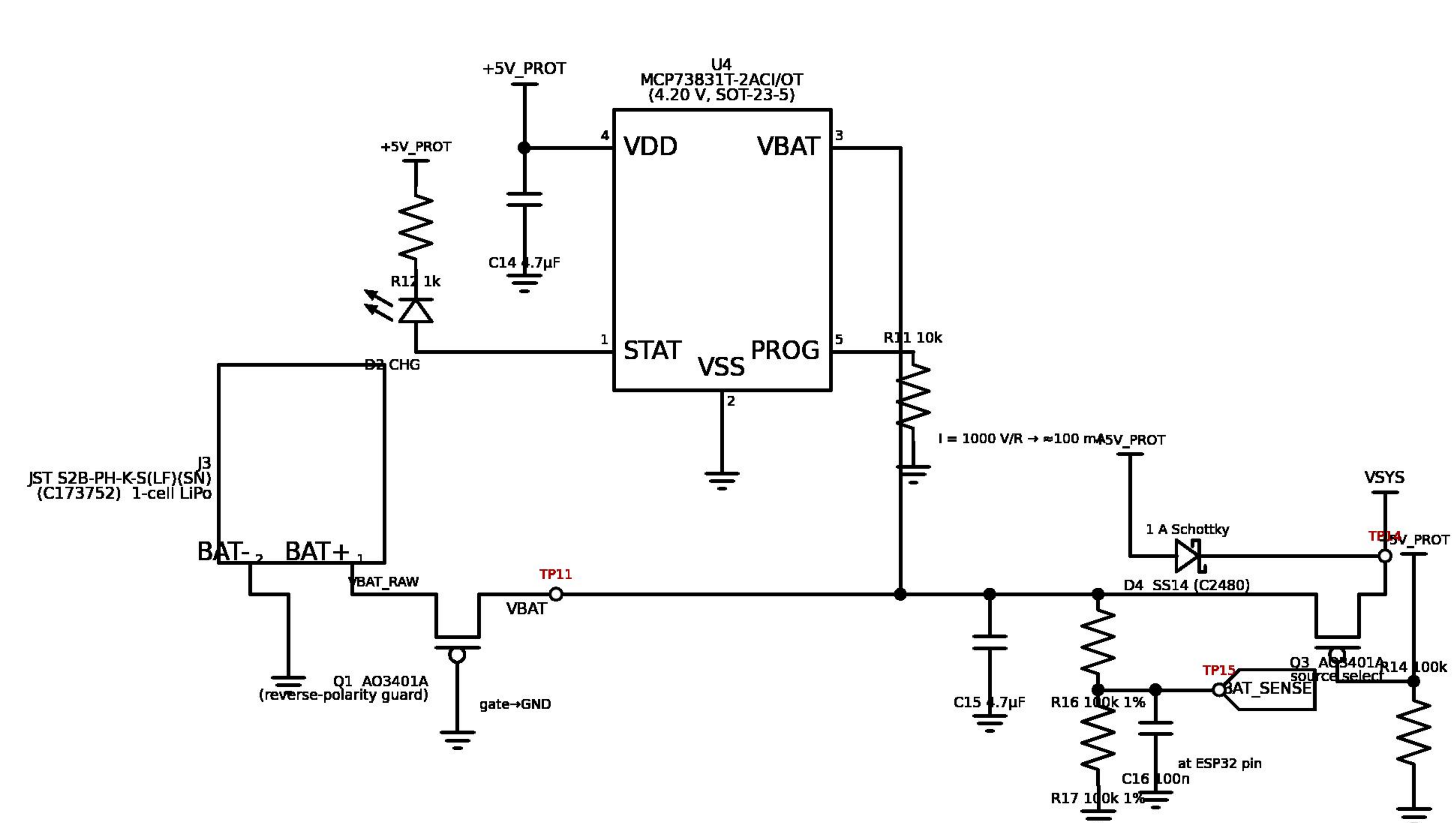
One pull-up pair serves the whole bus (90-day plan rule). CSB→VDDIO locks the BME280 into I²C mode; SDO must never float (→GND = addr 0x76). BME280 pin numbering runs CLOCKWISE in top view (DS §7.1) — check the footprint. Lid vent hole must stay uncovered; keep U5 away from LDO/module heat. U6 needs a clear light path; keep it away from status LEDs. Decoupling capacitors C10–C12 are drawn here, placed at the pins they name.

06 — SOIL PROBE INPUT & POWER SWITCH



SEN0193 output 0–3.0 V → 0–1.5 V at the pin (2:1); an accidental 5 V lands at 2.5 V — inside the ADC range (atten 3: 0–2900 mV). Q2 powers the probe only during readings (saves ≈5 mA continuous ≈ 120 mAh/day on battery); R15 holds it OFF through boot; firmware drives SENS_PWR_EN low to read. C13 placed at the ESP32 pin. Build-time: buzz out the probe cable against pads 1/2/3; if its far end is DuPont-style, use a PH-to-PH cable. Probe is NOT waterproof past its marked line.

07 — BATTERY, CHARGER & POWER PATH



Power path: USB present → D4 feeds VSYS (≈4.6 V); Q3's gate (5 V) sits ABOVE its source → Q3 OFF → battery isolated, charger owns it. USB removed → R14 pulls the gate to 0 V → Q3 ON → VSYS = battery, with the hand-over gap bridged by C1/C2/C4. Automatic — no firmware involved. Q1 blocks a reversed battery (body diode orientation + gate logic). R16/R17 halve VBAT → BAT_SENSE (2.1 V max at the ADC, ≈21 µA drain). D2: ON while charging, OFF when full; may flicker with no battery (STAT floats) — cosmetic. Battery: 1-cell protected LiPo ONLY; VERIFY plug polarity before first connect. Charger sleeps when USB is absent (internal reverse blocking).

NOTES

1. Net flags with identical names are connected across blocks: USB_DP, USB_DN, USB_VBUS, +5V_PROT, VSYS, +3V3, VBAT, BAT_SENSE, SENS_PWR_EN, ADC_SOIL, I2C_SCL, I2C_SDA. All grounds are one net (GND).
2. Block numbers match the planned KiCad hierarchical sheets 02–07 (01 = top sheet). Matches the Final Design Document §2–§9.
3. Open red circles = test points TP1–TP15. DNP = footprint only, Do Not Populate.
4. Battery power path: D4 + Q3 + R14 select USB or battery automatically; Q1 = reverse-polarity guard; Q2 = switched probe power; F1 upsized to 0.75 A hold; D3 TVS at the power entrance. Board runs from the battery when USB is absent.
5. Key parts: ESP32-S3-WROOM-1-N8 · AP2112K-3.3TRG1 · MCP73831T-2ACI/OT · USBLC6-2SC6 · AO3401A ×3 · SS14 · SMF5.0A · TYPE-C-31-M-12 · BME280 · VEML7700-TT · 1206L075/16WR · S3B-PH-K-S · S2B-PH-K-S.

Title	ESP32-S3 Plant Monitor — Reference Schematic
Project	RS_ESP32S3_PlantMonitor_RevA
Rev	A.2 (pre-capture reference drawing)
Date	2026-07-04 Sheet 1 of 1